



Effect of TiO₂ on B₂O₃-SiO₂-Al₂O₃-Na₂O-BaO glass lubricant: Viscosity, structure, melting behavior and wettability



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Abstract

Glass lubricants with suitable viscosity, softening temperature and wettability are critical for the high efficiency production of titanium alloys in hot extrusion process. A novel glass lubricant, (26.29–0.263x)B₂O₃–(41.19–0.412x)SiO₂–(6.13–0.061x)Al₂O₃–(11.39–0.114x)Na₂O–(15–0.15x)BaO–xTiO₂, was investigated in this paper. The effect of TiO₂ content on its viscosity, structure, melting behavior and wettability was investigated from 900°C to 1100°C. The results showed that the viscosity value at 950°C was decreased by 42.83% from 48.8Pas to 27.9Pas, when TiO₂ content was increased from 0wt% to 12wt%. The glass deformation, softening, hemispheric and flow temperatures were increased with an increased TiO₂ content, and the sample T2 (3wt% TiO₂) showed a good wettability with the Ti-6Al-4V alloy matrix. X-ray photoelectron spectroscopy fitting and molecular dynamics simulations due to show that the increased amounts of Q², Q³, Q⁴, [BO₃], [AlO₄], and [TiO₄] in the melts with an increased TiO₂

content lead to a weakened network homogeneity and symmetry, and consequently a reduced viscosity. Furthermore, lubricant T2 was recommended due to its softening temperature value of 759.1 °C, wetting angle value of 8.1° and viscosity value of 46.6Pas at 950 °C, and a small viscosity variation was observed from 950 °C to 1100 °C, which is a typical temperature change in TC4 hot extrusion.



Keywords

Glass lubricant; TiO₂ content; Melting behavior; Wettability; Viscosity; Structure

1. Introduction

Ti-6Al-4V alloy (TC4) is an $\alpha+\beta$ alloy with high specific strength and excellent corrosion resistance [1,2]. It is widely used in some areas, such as aerospace and biomechanics [3,4]. Hot extrusion is one of important methods for shaping titanium alloy products, with processing temperatures usually above 700 °C. This method facilitates manufacturing of metals with improved mechanical properties and fine microstructures [5,6]. In addition, the pressure range of hot extrusion is from 30 to 700MPa [7]. In order to improve die life, product surface quality and productivity, a glass lubricant is usually used for lubrication, thermal insulation and oxidation protection [8,9]. Among them, B₂O₃-SiO₂-based glass is considered as a proper lubricant for TC4 alloy hot extrusion from 950 °C to 1000 °C due to its adjustable viscosity and good thermal and chemical stability [[10], [11], [12]].

Viscosity, softening temperature, wettability, coefficients of friction, thermal expansion, thermal stability and conductivity are important properties of glass lubricants [[13], [14], [15]]. Among these properties, viscosity is one of the most important properties, which significantly affects its lubrication behavior [12,14,16]. Viscosity variation is influenced by melt composition and temperature, where the change of melt composition causes the evolution of melt network structure, affecting melt properties [[17], [18], [19]]. In recent years, the viscosity and structure relationship of B₂O₃-SiO₂-based melts have been widely studied by using X-ray photoelectron spectroscopy (XPS), Raman spectra, molecular dynamics (MD) simulation [12,[20], [21],

[22], [23], [24]]. Ni et al. [12] investigated the effect of B₂O₃/SiO₂ (B/Si) molar ratio and B₂O₃ content on viscosity and structure of B₂O₃-SiO₂-Al₂O₃-Na₂O-TiO₂ glass in the temperature range of 950°C–1100°C. The results showed that the viscosity value decreased when B/Si ratio was increased due to decreased amounts of BO, Q³, [BO₄] and [AlO₄]. It was also found that the viscosity was reduced with an increased B₂O₃ content, due to a decreased homogeneity and strength of melt network. Ke et al. [23] studied the effect of R₂O (R represents Na and K) on structure and viscosity of xR₂O-12.2B₂O₃-(86.7-x)SiO₂-1.1CaO fireproof glass in the temperature range of 1100°C–1550°C. The network degree of polymerization (DOP) and viscosity were found to be decreased as the molar ratios of Na₂O/SiO₂ and Na₂O/K₂O were increased. Fan et al. [24] used molecular dynamics (MD) simulations to study the effect of B₂O₃ content on structure and viscosity of B₂O₃-TiO₂-SiO₂-MgO-CaO slag at 1700°C. A reduced melt network strength as well as an increased diffusion coefficient were observed when B₂O₃ content was increased from 0wt% to 24wt%. This led to a gradually decreased viscosity. The above studies have demonstrated the feasibility of XPS and Raman spectra, MD simulation to investigate the viscosity change mechanism of B₂O₃-SiO₂-based glass melts.

The softening temperature (T_s) of glass lubricant is also an important property [16,[25], [26], [27]]. A proper softening temperature ensures that the lubricant can soften and quickly cover the billet surface to protect billet from oxidation protection [16,25]. Investigating the melting or sintering behavior of the melt using goniometric techniques allows to determine melt T_s [16,25]. Zhang et al. [30] investigated the effect of SiO₂/BaO (Si/Ba) mass ratio on the sintering behavior of (30.5-x)BaO-(16+x)SiO₂-15B₂O₃-15SrO-2Al₂O₃-9.5ZnO-6TiO₂-3MgO-3Na₂O glass-ceramics in the temperature range of 600°C–1150°C. It was found that the sintering temperature (including softening, hemispherical and flow temperatures) was increased by about 20°C with an increased Si/Ba ratio from 0.5 to 0.9. This is caused by both the increased Si/Ba ratio that promotes the translation of [BO₃] to [BO₄], and the formation of more Si-O-B bonds by [BO₄] connected to [SiO₄]. He et al. [34] studied the effect of TiO₂/B₂O₃ (Ti/B) mass ratio on sintering behavior, crystallization behavior and structure of 27.5BaO-19SiO₂-15SrO-9.5ZnO-xTiO₂-(21-x)B₂O₃-based glass-ceramics from 550°C to 1070°C. It was found that the glass transition and sintering temperatures were increased when Ti/B ratio was increased. This was caused by an increased network DOP due to increased amounts of [BO₄] and [TiO₄]. In addition, previous studies [35,36] have shown that the melt softening temperature can be determined using a goniometer, which provides the wetting angle (θ) between the melt and the matrix. This characterizes the wettability between melt and matrix. Among them, $\theta < 90^\circ$ indicates that the melt may be a wetting matrix, and a smaller θ represents a better wettability [36]. Therefore, a melt with good wettability can adhere to the matrix. However, the softening temperature and wettability of B₂O₃-SiO₂-based melts in the hot extension working temperature range, e.g. 750 °C–1000°C, were not investigated [[29], [30], [31], [32], [33], [34], [35], [36]].

The above melt properties are influenced by the structure evolution due to the change of melt composition [17,18,34]. Previous studies have found that B₂O₃ and SiO₂ acted as network formers, and Na₂O and BaO acted as network modifiers, while Al₂O₃ and TiO₂ acted as network intermediates [34,[37], [38], [39]]. Among them, the addition of TiO₂ can improve the network strength, thermal and chemical stability of B₂O₃-SiO₂-based melts [17,34]. However, a further increased TiO₂ content weakens the network homogeneity and symmetry, which has an impact on the melt properties [40]. In addition, previous studies regarding the effect of TiO₂ on the structure and properties of B₂O₃-SiO₂-based glass mainly focus on the crystallization behavior, phase separation, and glass stability [41,42]. Kwindt et al. [42] investigated the effect of TiO₂ content on crystallization, phase separation and thermal stability of 7Na₂O-23B₂O₃-(70-x)SiO₂-xTiO₂ glass in the temperature range of 630°C–740°C. It was found that no influence on the glass thermal stability and the formation of sodium borate-rich phases during phase separation was observed with an increased TiO₂ content from 0mol.% to 14.6mol.%. Also, the simultaneous occurrence of phase separation and crystallization leads to a slow mass transfer, resulting in slightly lower domain sizes of microphases in TiO₂-containing glass compared to TiO₂-free glass. However, the effect TiO₂ on its viscosity, structure, softening temperature and wettability has rarely been reported. Additionally, when a B₂O₃-SiO₂-based lubricant adheres to the surface of a titanium alloy, B in the melt reacts slightly with Ti element in alloy, which may affect the surface quality of the titanium alloy product [14,43]. Adding TiO₂ can reduce B₂O₃ content in the melt while avoiding the reaction between the new components and Ti elements in the alloy, weakening the above reaction and obtaining titanium alloys with better surface quality. In order to determine the appropriate TiO₂ addition amount to obtain good properties of B₂O₃-SiO₂-based glass lubricants. The effect of TiO₂ on the viscosity, melt structure, softening temperature and wettability of (26.29–0.263x)B₂O₃-(41.19–0.412x)SiO₂-(6.13–0.061x)Al₂O₃-(11.39–0.114x)Na₂O-(15–0.15x)BaO-xTiO₂ (BSANBT) glass was investigated in this paper. Since B-, Ti- and Al-related units can also be present in different forms in oxide melts, a detailed analysis of the evolution of Si-, B-, Ti- and Al-related units was studied by using both XPS and MD simulations.

2. Experimental and simulation methods

2.1. Experiments

The target glass lubricant samples were prepared by analytical reagents from Shanghai Macklin Biochemical Co., specifically SiO₂ (≥99wt%), B₂O₃ (≥98wt%), Al₂O₃ (≥99wt%), TiO₂ (≥99wt%), Na₂CO₃ (≥99.8wt%) and BaCO₃ (≥99wt%). TiO₂ contents in the designed glass lubricants are 0wt%, 3wt%, 6wt%, 9wt% and 12wt%, as listed in [Table 1](#). For each sample, 130g oxide powders were weighed

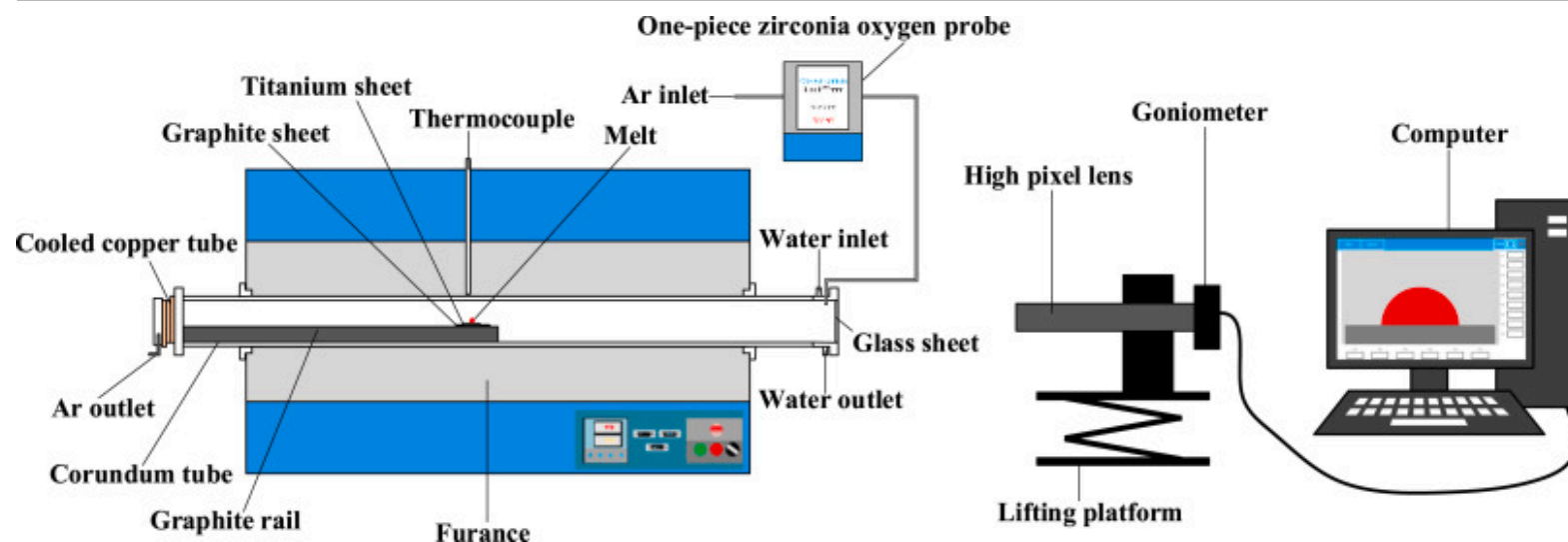
and homogeneously mixed. Then, it was pre-melted in a graphite crucible at 1300°C in an air atmosphere for 60min to homogenize the melt. Finally, the melted glass was quenched in a copper pot with a water bath underneath. In addition, the pre-melted glass was held at 950°C for 30min, after which it was quenched to room temperature. It was crushed and sieved through a 200 mesh sieve, and its structure was investigated by X-ray photoelectron spectroscopy (XPS, Thermo ESCALAB 250XI, America) and X-ray diffraction (XRD, Shimadzu XRD-7000, Japan).

Table 1. Chemical compositions of the studied BSANBT glass.

No.	Mixed glass (wt.%)						Mixed glass (mol.%)					
	B ₂ O ₃	SiO ₂	Al ₂ O ₃	Na ₂ O	BaO	TiO ₂	B ₂ O ₃	SiO ₂	Al ₂ O ₃	Na ₂ O	BaO	TiO ₂
T1	26.29	41.19	6.13	11.39	15.00	0	26.88	48.80	4.28	13.08	6.96	0
T2	25.50	39.95	5.95	11.05	14.55	3.00	26.16	47.49	4.16	12.73	6.78	2.68
T3	24.71	38.72	5.76	10.71	14.10	6.00	25.43	46.17	4.05	12.38	6.59	5.38
T4	23.92	37.48	5.59	10.36	13.65	9.00	24.70	44.84	3.93	12.02	6.40	8.10
T5	23.14	36.25	5.39	10.02	13.20	12.00	23.97	43.51	3.82	11.66	6.21	10.84

Glass viscosity was measured by a viscometer (RTW-10, Northeastern University, China) based on the rotating cylinder method [12]. The details of the viscometer measurement are introduced in our previous study [12]. The glass softening temperature was determined by measuring the glass melting behavior using a goniometer. The goniometer can also analyze the wettability of glass lubricant on TC4 alloy matrix. In this study, the goniometer was a custom equipment, as shown in Fig. 1. It was mainly consisted of a vacuum furnace (OTF-1200X, Hefei Ke Jing Materials Technology CO., LTD, China), a high pixel lens (DH-HV1351UM, DAHENG Imaging, China) and a one-piece zirconia oxygen probe (SP-0ACT1, Senspro, China). First, the sieved glass powder was weighed about 0.1 g and pressed at 10MPa to form a cylinder sample with a diameter of 4mm and a height of 4mm. In addition, TC4 titanium sheet with a length of 23.5mm, a width of 17.5mm and a height of 3mm was smoothly polished using 400-, 800- and 1200-grit silicon carbide sandpaper and diamond polishing agent. Then, the sample was placed in the center of the titanium sheet and slowly moved into the furnace hot zone. High-purity Ar gas (partial oxygen pressure of $3.94564 \times 10^{-}$

¹⁴Pa), treated by heating titanium sheet at 700°C, was passed through the furnace for 90 min before preheating. Finally, the sample was heated, and a goniometer was used to photograph and record the sample melting behavior. Additionally, the melting behavior was also investigated using differential scanning calorimetry/thermogravimetric analysis (DSC/TGA, SDT Q600 SPECIFICATIONS, America).



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Fig. 1. Experimental equipment for glass melting behavior measurement.

2.2. Simulations

Molecular dynamics (MD) simulation of melt structure was carried out by using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [44]. With the use of periodic boundary conditions, a random distribution of about 6000 atoms was modeled. Among them, the atomic number, density and side length of the three-dimensional cubic box are shown in Table 2.

Table 2. Atomic number, box length and density of the studied BSANB glass.

No.	B	Si	Al	Na	Ba	Ti	O	Total	Density/g·cm ⁻³	Box length/Å
T1	908	824	146	442	118	0	3568	6006	3.03	40.4
T2	888	806	142	432	115	46	3580	6009	3.02	40.6
T3	868	787	138	422	113	92	3591	6011	3.03	40.6
T4	846	767	136	412	110	139	3601	6011	3.03	40.7
T5	824	748	132	402	107	187	3612	6012	3.06	40.7

The glass densities were calculated by a reasonable assumption using the oxide densities and the mass fraction of the components. In recent years, the Buckingham potential has been found to be good at simulating the structure and properties of silicates and borosilicates [40,[45], [46], [47]]. Furthermore, the Buckingham potential comprised the short-range exponential repulsive term, the long-range Coulombic interactions and the van der Waals force potential. The potential function is calculated as follows:

$$U_{ij} = \frac{q_i q_j}{r_{ij}} + A_{ij} \exp\left(-\frac{r_{ij}}{\rho_{ij}}\right) - \frac{C_{ij}}{r_{ij}^6} \quad (1)$$

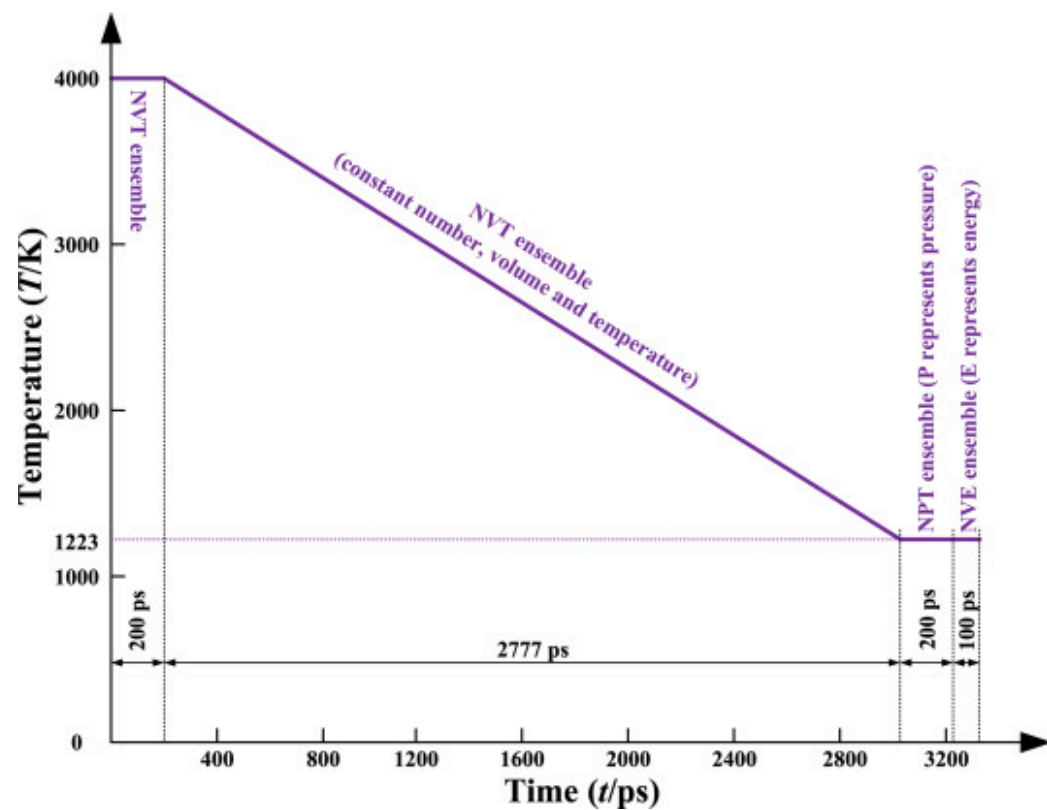
where U_{ij} is the interatomic potential of atoms i and j in kJ·mol⁻¹, q_i and q_j are the charges of atoms i and j in e, A_{ij} and C_{ij} are energy parameters describing the repulsive force energy and the dispersive force, respectively, r_{ij} is the distance between atoms i and j in Å and ρ_{ij} is the distance parameter between atoms i and j . A_{ij} , ρ_{ij} and C_{ij} are detailed in [Table 3](#).

Table 3. Buckingham potential parameters used in this simulation [40,[46], [47], [48]].

Bond and partial charge (e)	A_{ij} (kJ·mol ⁻¹)	ρ_{ij} (Å)	C_{ij} (Å ⁶ kJmol ⁻¹)
O ^{-1.2} -O ^{-1.2}	2029.2204	0.343645	192.58
B ^{1.8} -O ^{-1.2}	12648.5242	0.171271	28.500
Si ^{2.4} -O ^{-1.2}	13702.9050	0.193817	54.681

.Bond and partial charge (e)	A_{ij} (kJ·mol⁻¹)	ρ_{ij} (Å)	C_{ij} (Å⁶kJmol⁻¹)
Al ^{1.8} -O ^{-1.2}	12201.4170	0.195628	31.997
Ti ^{2.4} -O ^{-1.2}	23707.895	0.185563	14.513
Na ^{0.6} -O ^{-1.2}	4383.756	0.243838	30.7
Ba ^{1.2} -O ^{-1.2}	1214.102	0.352237	8

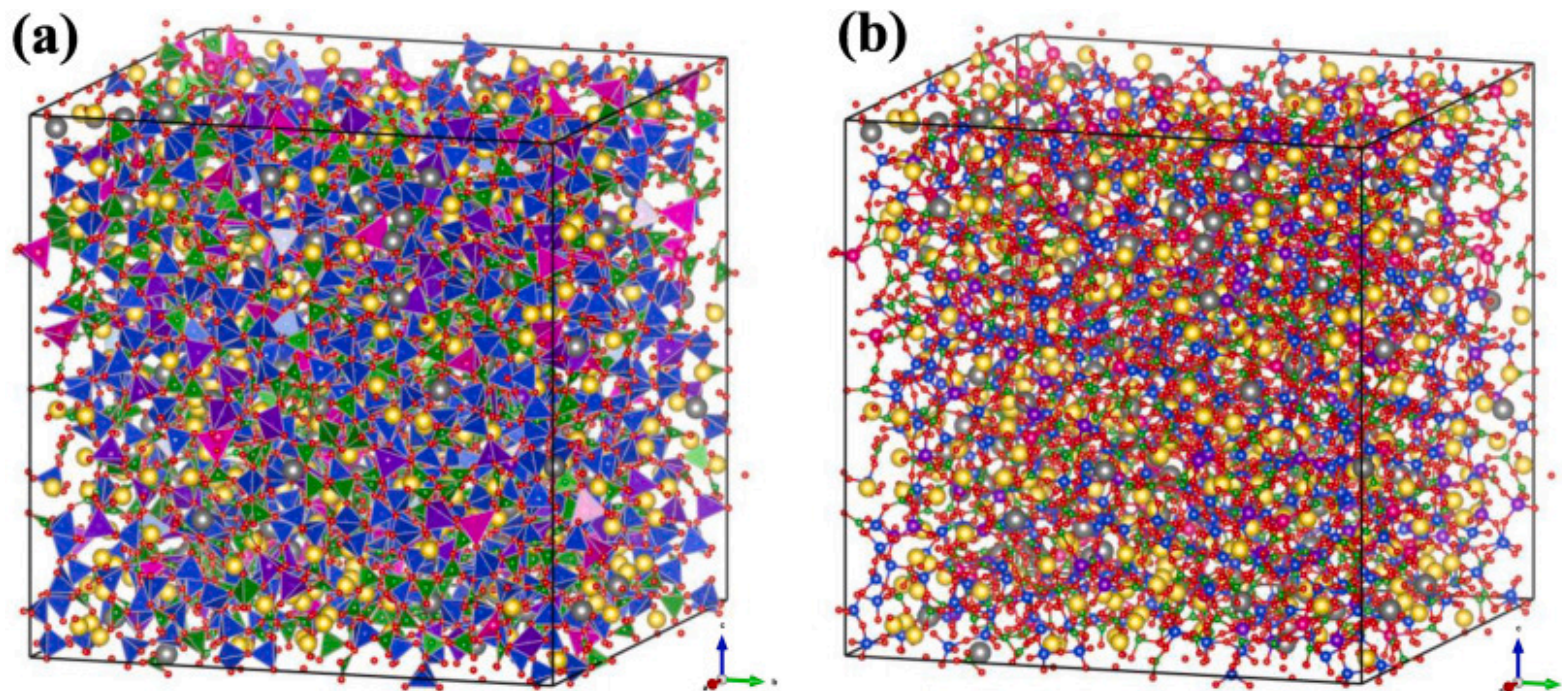
The long-range Coulomb force was evaluated by using the Ewald sum method in the simulation, and the cut-off radius for the short-range repulsive force was set to 10Å. The motion equations were integrated by using the Verlet Leapfrog algorithm with a time step of 1 fs. The melting process of glass from 4000K to 1223K is calculated using different ensembles, as shown in [Fig. 2](#). The melt structure was analyzed by using VESTA (Visualization for Electronic and Structural Analysis) and MATLAB [49] after the calculation. The complex network of B-, Si-, Al-, and Ti- related structural unit connections and the various atomic distributions in the BSANBT glass are shown in [Fig. 3](#).



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Fig. 2. MD simulation of BSANBT glass melting process using ensemble and time.



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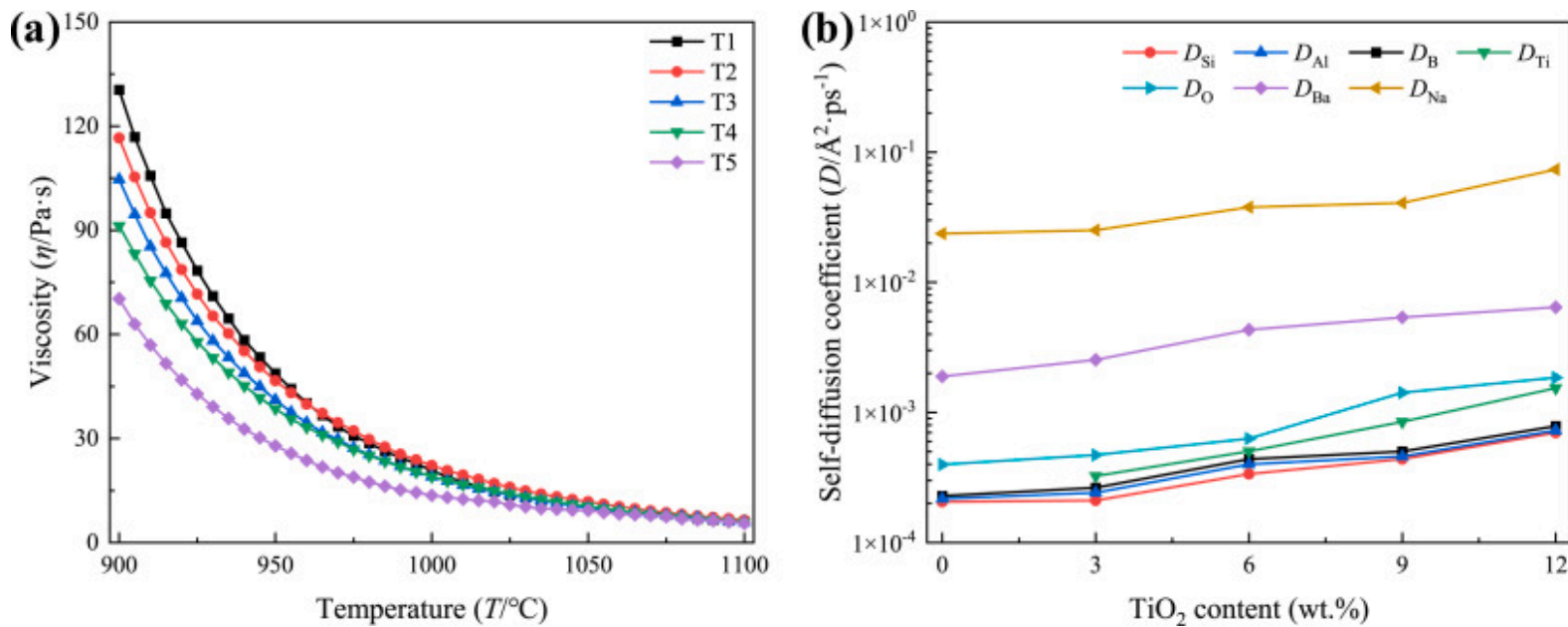
Fig. 3. Snapshot of MD simulated for sample T3 structure using VESTA: (a) Full view of the simulation cell, (b) Enlarged snapshot showing detailed glass structure. The colored balls are oxygen (red), silicon (blue), aluminum (purple), titanium (pink), boron (green), sodium (yellow) and barium (grey), respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3. Results and discussion

3.1. Effect of TiO₂ content on melt viscosity

Fig. 4 shows the viscosity and atomic diffusion coefficient variations with different TiO₂ contents. The viscosity shows a roughly decreased trend with an increased TiO₂ content, as shown in Fig. 4a. The viscosity value at 950°C was changed from 48.8Pas to 27.9Pas, decreased by 42.83%, when TiO₂ content was increased from 0wt% to 12wt%. Also, the viscosity values at 1000°C with

TiO₂ content of 0wt%, 3wt%, 6wt%, 9wt% and 12wt% were reduced by 57.38%, 52.36%, 53.53%, 50.78% and 51.25%, respectively, compared to those at 950°C viscosity value. This indicated a small variation of the sample viscosity as a function of temperature from 950°C to 1000°C.



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Fig. 4. Effect of TiO₂ content on the variations of viscosity and atomic self-diffusion coefficient: (a) viscosity, (b) atomic self-diffusion coefficient at 950°C.

The relationship between the viscosity η and the atomic self-diffusion coefficient D can be described by the Stokes-Einstein equation [24,45]. The calculated equation between D and η are as follows:

$$D = \frac{kT}{3\pi\lambda\eta} \quad (2)$$

where T is temperature in K, k is Boltzmann constant in m²·kg·s⁻²·K⁻¹, λ is jumping distance for diffusion in Å. Here λ was usually set to the diameter of one oxygen atom for a molten oxide system [50]. Assuming that each atom in the melt is an individual particle, the average diffusivity can be quantized by calculating the mean square displacement (MSD, $\Delta^2 r(t)$), which in turn

allows the calculation of D [45]. D and MSD of a single particle are calculated as follows:

$$D = \lim_{x \rightarrow \infty} \frac{1}{6} \cdot \frac{d\Delta^2 r(t)}{dt} \quad (3)$$

$$\Delta^2 r(t) = \frac{1}{N} \sum_{i=1}^N (r_i(t) - r_i(0))^2 \quad (4)$$

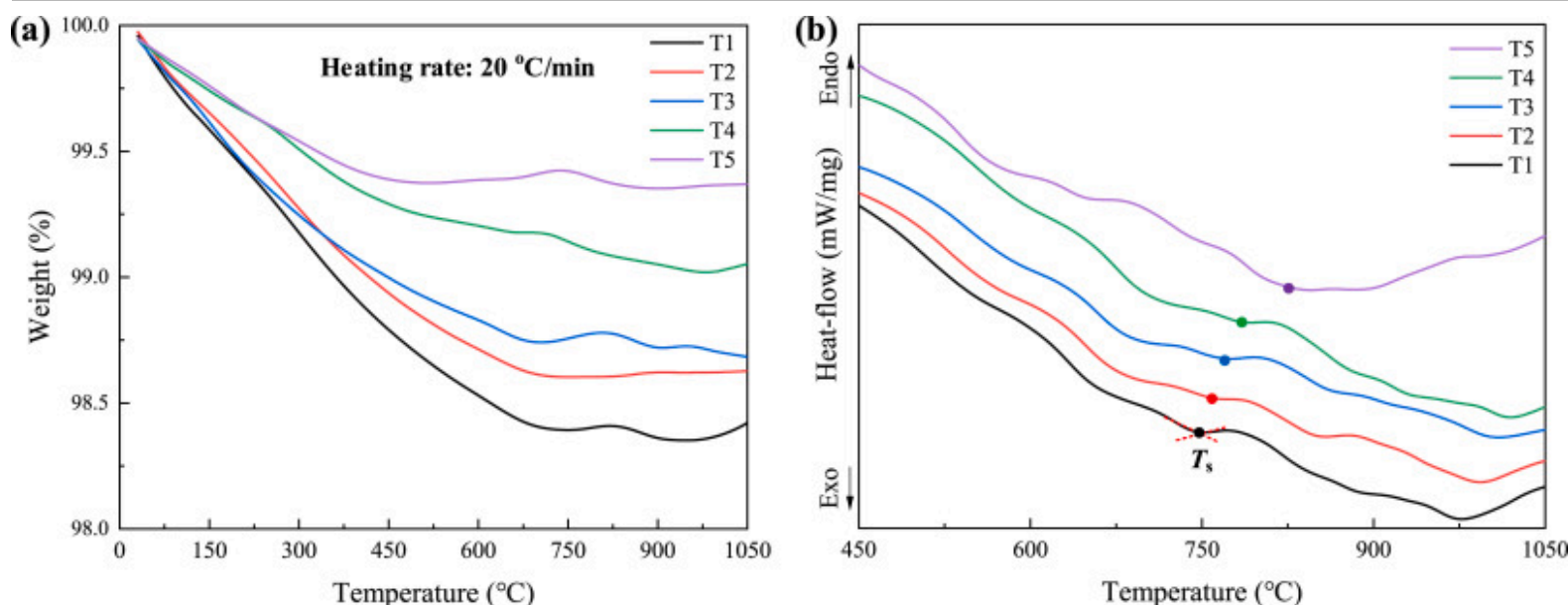
where N is the total number of measured particles, $r_i(t)$ and $r_i(0)$ are the positions of atom i at observation time t and at the initial moment, respectively.

Fig. 4b Shows that the atomic self-diffusion coefficients of B, Si, Al, Ti, Na, Ba and O increased with an increased TiO₂ content at 950°C. The correlation between the atomic self-diffusion coefficients is as follows: $D_{\text{Si}} < D_{\text{Al}} < D_{\text{B}} < D_{\text{Ti}} < D_{\text{O}} < D_{\text{Ba}} < D_{\text{Na}}$. Furthermore, the diffusion velocities of Na and Ba are about 100 and 10 times that of the other five atoms, respectively. This means that the atoms of Na and Ba are not involved in the complex network composed of B, Si, Al, and Ti related units, and can freely move within the network. The magnitude relationships between D_{Si} , D_{Al} , D_{B} and D_{Ti} also indicate that the bond energy of Si-O is larger than that of Al-O in BSANBT glass. Furthermore, the bond energy of Al-O is larger than that of B-O, and the bond energy of Ti-O is the smallest among them. This is due to the fact that Si atoms form stable [SiO₄], while B, Al, and Ti Al-related units can also be present in various structures in the glass. This leads to a minimum self-diffusion coefficient for Si atoms. Additionally, the atom with the minimum self-diffusion coefficient directly correlates with its rheological behavior and determines the melt viscosity [24,50]. This means that Si atomic self-diffusion coefficient at 950°C is increased with an increased TiO₂ content, which also verifies the variation of BSANBT glass viscosity with an increased TiO₂ content.

3.2. Effect of TiO₂ content on melting behavior and wettability between melt and TC4 alloy

Fig. 5 shows the TG and DSC curves of BSANBT glass lubricant. According to the TG curves in Fig. 5a, it was found that the lubricant has less than 2% weight loss between 30°C and 650°C, and the weight loss does not change much at temperatures greater than 650°C. This may be caused by the volatilization of B₂O₃ [14]. It was also found that the weight loss of the lubricant was reduced with an increased TiO₂ content, which indicates that adding TiO₂ to the BSANBT lubricant can improve its thermal stability. In Fig. 5b, the softening temperature (T_s) of lubricants T1, T2, T3 and T4 is found to be in the range of 750.7°C–787.4°C, while the T_s value for lubricant T5 is 820.6°C. This is because an increased TiO₂ content resulted in reduced B₂O₃, BaO and Na₂O contents in the melt, which have a fluxing effect, leading to an increased T_s value [34,51]. Additionally, the T_s values of lubricants

T1, T2, T3 and T4 are much lower than the TC4 alloy hot extrusion working temperature (950°C). This means the lubricant forms a viscous coating that adheres to the metal surface during hot extrusion, providing effective oxidation protection.



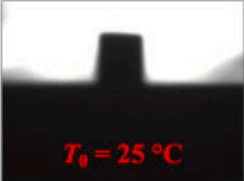
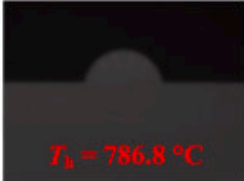
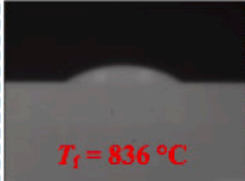
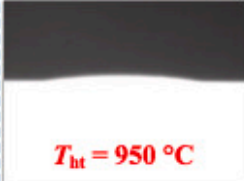
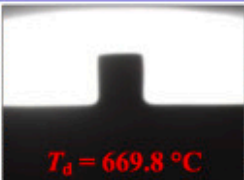
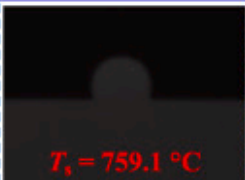
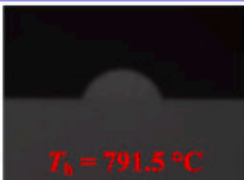
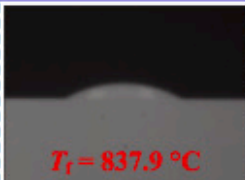
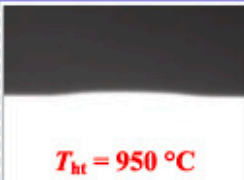
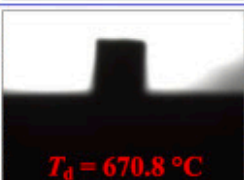
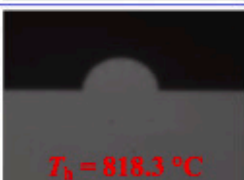
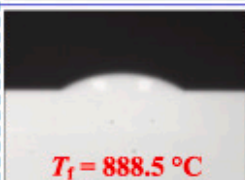
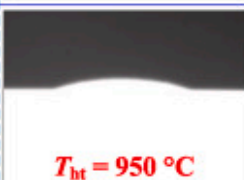
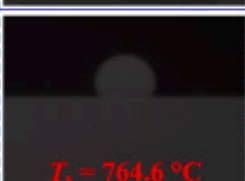
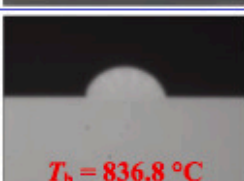
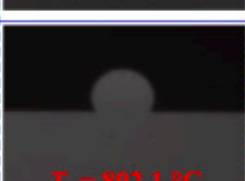
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Fig. 5. TG and DSC curves of the BSANBT glass lubricants: (a) TG and (b) DSC.

Fig. 6 shows the variation of the melting behavior of BSANBT glass with an increased TiO₂ content. Among them, an initial sample temperature at which deformation occurs with an increased temperature is the deformation temperature (T_d), and the temperatures corresponding to 75%, 50% and 25% of initial sample height are T_s , hemispherical temperature (T_h) and flow temperature (T_f), respectively. Also, a smaller difference between T_s , T_h and T_f for the melt indicates a better melt flow index [34,51,52]. The results showed that T_d , T_s , T_h and T_f were increased by 1.4°C, 8.2°C, 4.7°C and 11.9°C, respectively, when TiO₂ content was increased from 0wt% to 3wt%. With a further increased TiO₂ content to 12wt%, T_d , T_s , T_h and T_f values were increased by 30.2°C, 43°C, 89.8°C and 201.6°C, respectively. Of these, the T_s values of the lubricant in Fig. 6 differ by $\pm 2.3\%$ from those in Fig. 5b, confirming the accuracy of the observed melting behavior of the lubricant using a goniometer. An analysis of the subsequent structural evolution revealed that the increased amounts of BO and [TiO₄] suggest that the network gradually polymerizes when TiO₂ content is increased. This indicates that more Ti⁴⁺ ions are entering the melt network, leading to a

gradual increased T_s , T_h and T_f values during lubricant melting process. In addition, when lubricant has low T_h and T_f , it also indicates that its rapid melting allows the formation of a molten film on the titanium billet. This indicates that samples T1, T2 and T3 fulfill the requirements of T_s , T_h and T_f for the lubricant in the TC4 alloy hot extrusion from 950 °C to 1000 °C.

Initial temperature	Deformation temperature	Softening temperature	Hemispheric temperature	Flow temperature	Holding temperature
(a)  $T_0 = 25\text{ }^{\circ}\text{C}$	 $T_d = 668.4\text{ }^{\circ}\text{C}$	 $T_s = 750.9\text{ }^{\circ}\text{C}$	 $T_h = 786.8\text{ }^{\circ}\text{C}$	 $T_f = 836\text{ }^{\circ}\text{C}$	 $T_{ht} = 950\text{ }^{\circ}\text{C}$
(b)  $T_0 = 25\text{ }^{\circ}\text{C}$	 $T_d = 669.8\text{ }^{\circ}\text{C}$	 $T_s = 759.1\text{ }^{\circ}\text{C}$	 $T_h = 791.5\text{ }^{\circ}\text{C}$	 $T_f = 837.9\text{ }^{\circ}\text{C}$	 $T_{ht} = 950\text{ }^{\circ}\text{C}$
(c)  $T_0 = 25\text{ }^{\circ}\text{C}$	 $T_d = 670.8\text{ }^{\circ}\text{C}$	 $T_s = 763.9\text{ }^{\circ}\text{C}$	 $T_h = 818.3\text{ }^{\circ}\text{C}$	 $T_f = 888.5\text{ }^{\circ}\text{C}$	 $T_{ht} = 950\text{ }^{\circ}\text{C}$
(d)  $T_0 = 25\text{ }^{\circ}\text{C}$	 $T_d = 686.2\text{ }^{\circ}\text{C}$	 $T_s = 764.6\text{ }^{\circ}\text{C}$	 $T_h = 836.8\text{ }^{\circ}\text{C}$	 $T_f = 985.4\text{ }^{\circ}\text{C}$	 $T_{ht} = 1000\text{ }^{\circ}\text{C}$
(e)  $T_0 = 25\text{ }^{\circ}\text{C}$	 $T_d = 700\text{ }^{\circ}\text{C}$	 $T_s = 802.1\text{ }^{\circ}\text{C}$	 $T_h = 881.3\text{ }^{\circ}\text{C}$	 $T_f = 1039.5\text{ }^{\circ}\text{C}$	 $T_{ht} = 1050\text{ }^{\circ}\text{C}$

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Fig. 6. Effect of TiO₂ content on melting behavior of BSANBT lubricant: (a) T1 (0wt% TiO₂), (b) T2 (3wt% TiO₂), (c) T3 (6wt% TiO₂), (d) T4 (9wt% TiO₂) and (e) T5 (12wt% TiO₂).

Wetting angle (θ) is a parameter which characterizes the wettability between melt and matrix. The melt and matrix are completely unwetted when θ is 180°. The melt and matrix are unwetted when θ is in the range of 90°–180°. The melt and matrix are wetted when θ is in the range of 0°–90°. Furthermore, the melt and matrix are completely wetted when θ is 0° [53,54]. The angle at the interface between lubricant melt and TC4 titanium alloy was measured using Image J software and the results are listed in Table 4. When the temperature reached T_f and holding temperature (T_{ht}), the wetting angle between the lubricant and TC4 alloy was first decreased with TiO₂ content increased from 0wt% to 3wt%, and then it was increased with a further increased TiO₂ content to 12wt%. This may be caused by reactive wettability between glass lubricant and matrix with a further increased TiO₂ content [36,55]. Furthermore, the contact angle between BSANBT lubricant and TC4 alloy was in the range of 8.1°–32.4° when temperature was increased from T_f to T_h .

Table 4. Wetting angle between BSANBT lubricant and TC4 alloy at different temperatures.

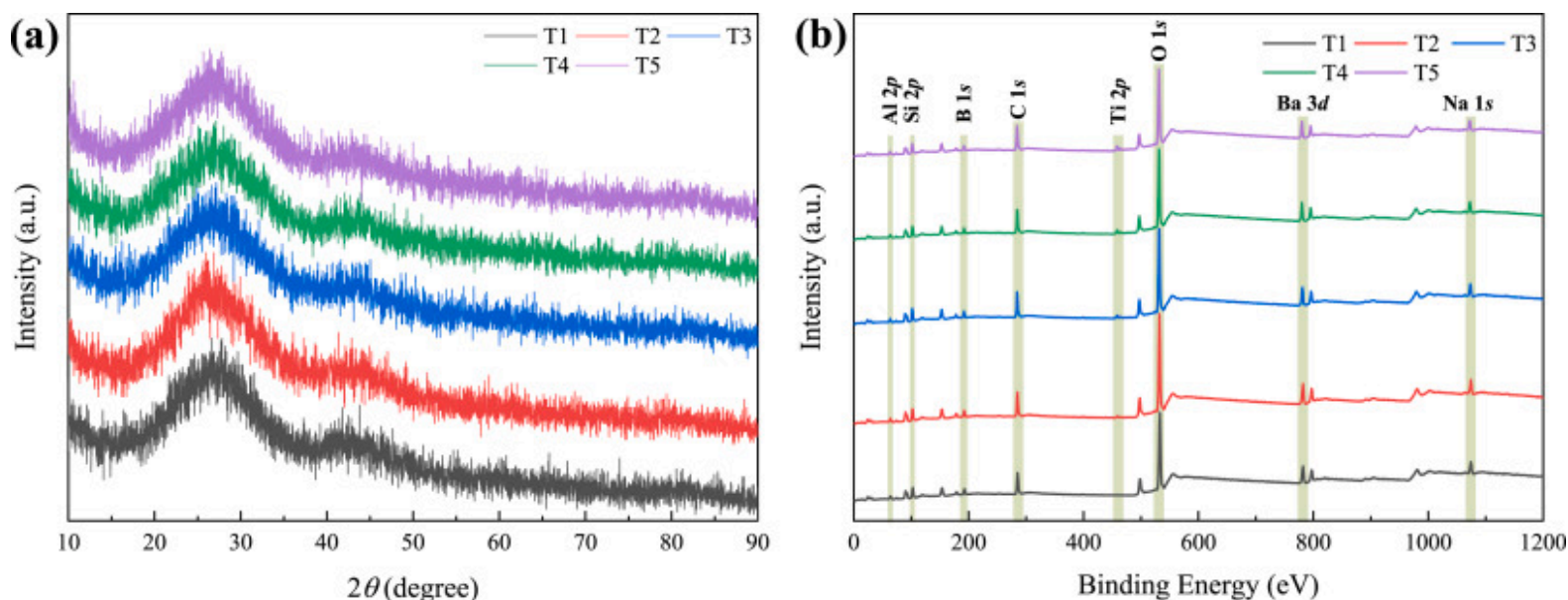
TiO ₂ content	T1 (0wt%)	T2 (3wt%)	T3 (6wt%)	T4 (9wt%)	T5 (12wt%)
T_f	32.4°	25.3°	27.8°	28.7°	29.4°
T_{ht}	9.5°	8.1°	16.0°	20.9°	22.4°

3.3. Effect of TiO₂ content on melt structure

3.3.1. XPS spectral analysis

Fig. 7 shows the XRD and XPS spectral curves of BSANBT glass with different TiO₂ contents. Fig. 7a displays the XRD results of the BSANBT glass, and it can be seen that no obvious crystal diffraction peaks were found for the pre-melted glass. This means that the quenching sample was in an amorphous state. XPS analysis was carried out to reveal the evolution of Si-, B-, Al- and Ti-related units in the glass melt with different TiO₂ content. The C 1s peak at 284.8eV was used as the binding energy for correction. Fig. 7b illustrates 0–1200eV peaks produced by the constituent elements of BASNBT glass in the Al 2p, Si 2p, B 1s, Ti

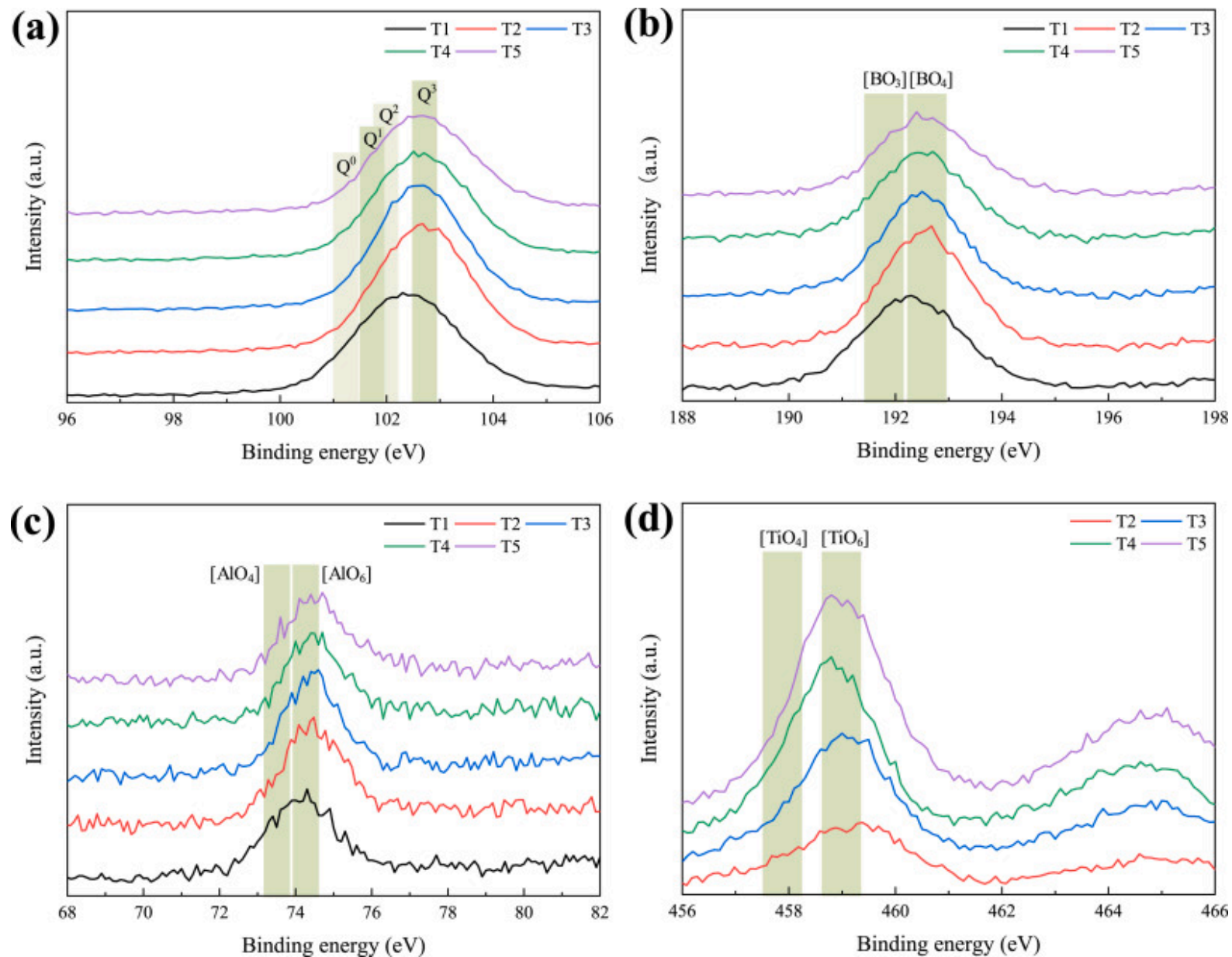
2p, Ba 3d, O 1s and Na 1s regions. The peaks of Si 2p, B 1s, Al 2p and Ti 2p were scanned to obtain high-resolution core-level spectra, as shown in Fig. 8.



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Fig. 7. XRD and XPS spectral curves of glass with different TiO₂ contents: (a) XRD and (b) XPS.

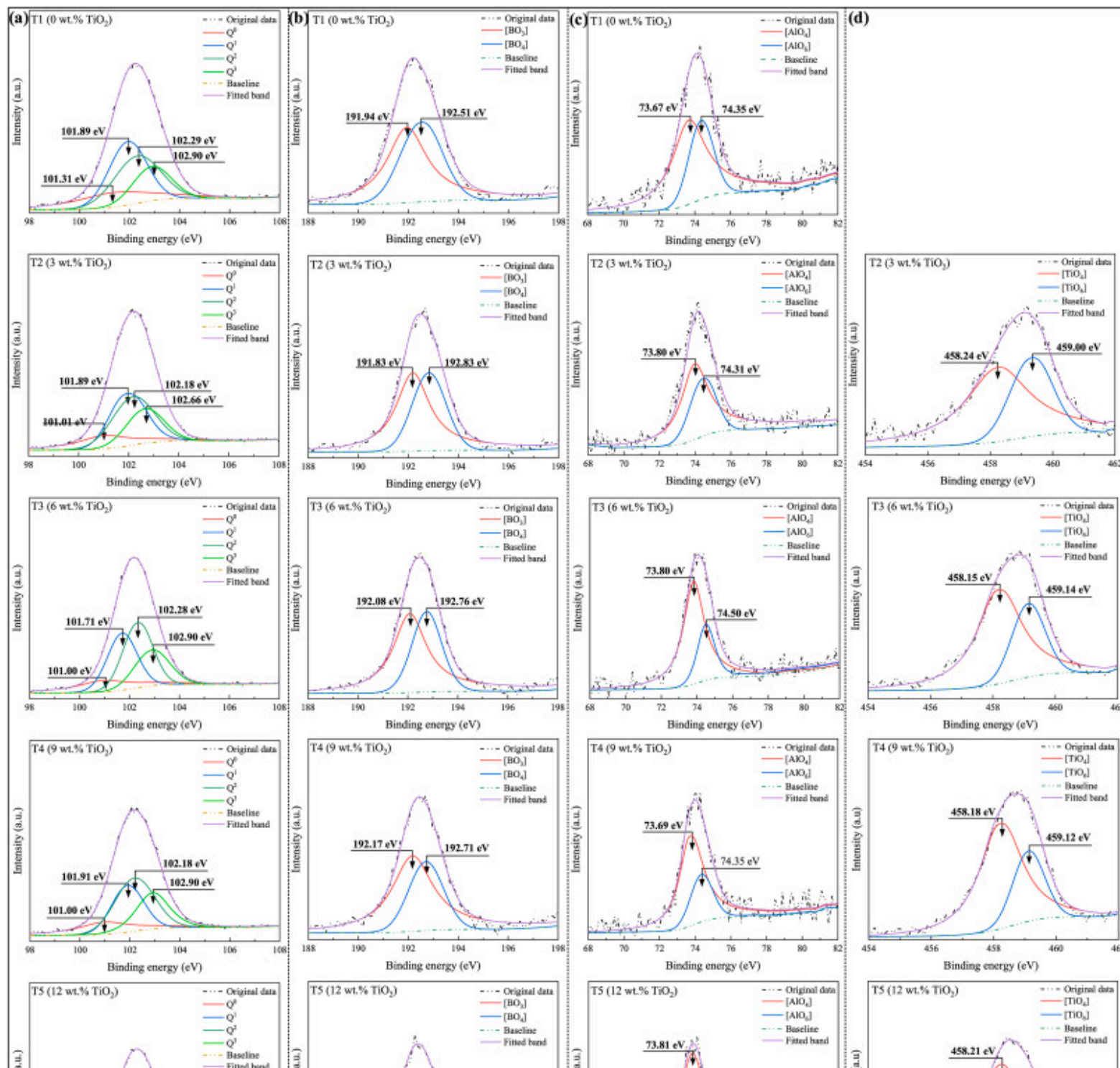


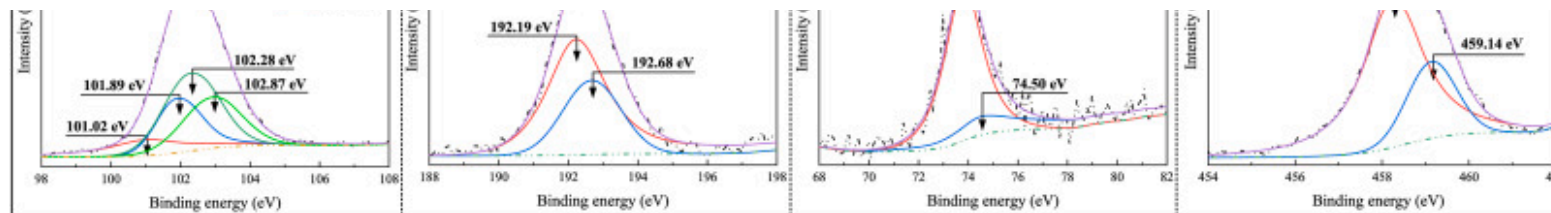
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Fig. 8. High-resolution core-level XPS spectral profile of BSANBT glass with various TiO₂ content: (a) Si 2*p*, (b) B 1*s*, (c) Al 2*p* and (d) Ti 2*p*_{3/2}.

The Si-related unit in the melt is composed of [SiO₄], which is denoted by Q^{*n*}, with *n* representing the number of BO for each unit. According to previous studies [22,[56], [57], [58], [59], [60], [61], [62], [63]], the XPS binding energies were at approximately 101.3±0.3 eV for Q⁰ [56,57], 101.6±0.3 eV for Q¹ [[56], [57], [58]], 102.0±0.3 eV for Q² [[56], [57], [58]], 102.6±0.3 eV for Q³ [[56], [57], [58]], 191.9±0.3 eV for [BO₃] [60,61], 192.6±0.3 eV for [BO₄] [60,61], 73.5±0.3 eV for [AlO₄] [22,59], 74.2±0.3 eV for [AlO₆] [22,59], 457.9±0.3 eV for [TiO₄] [62,63] and 459±0.3 eV for [TiO₆] [62,63]. Among them, XPS signal of Ti 2*p* includes two peaks: Ti 2*p*_{1/2} and Ti 2*p*_{3/2}. Ti 2*p*_{3/2} with a higher signal intensity is generally used for Ti coordination analysis [62]. The fitted Gauss-Lorentz curves for specific Si-, B-, Al- and Ti-related unit fitting are shown in Fig. 9, and its peak area fraction can be calculated, as shown in Fig. 10.

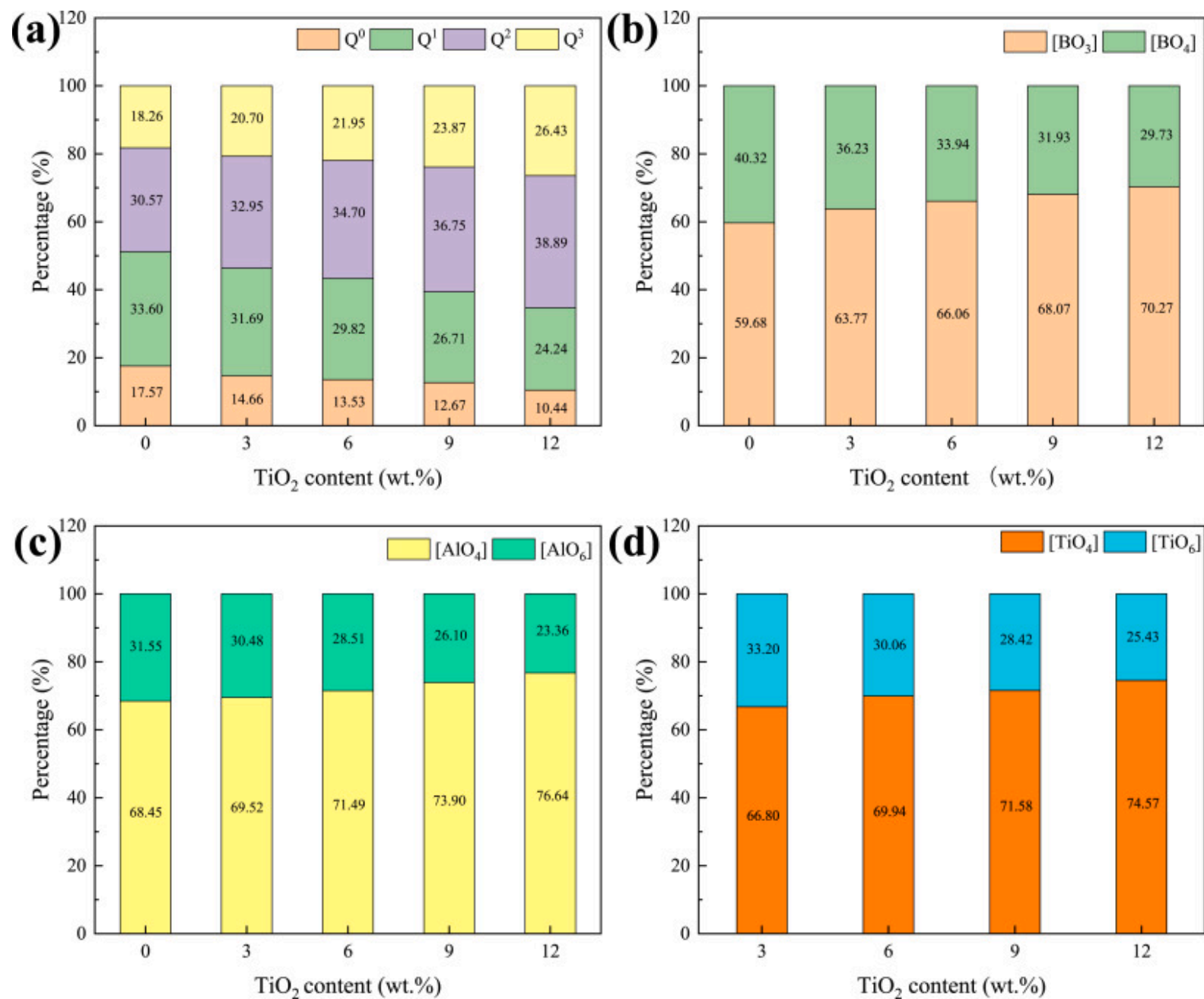




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Fig. 9. Deconvolution results of XPS spectra for BSANBT glass at 950°C: (a) Si 2p, (b) B 1s, (c) Al 2p and (d) Ti 2p_{3/2}.



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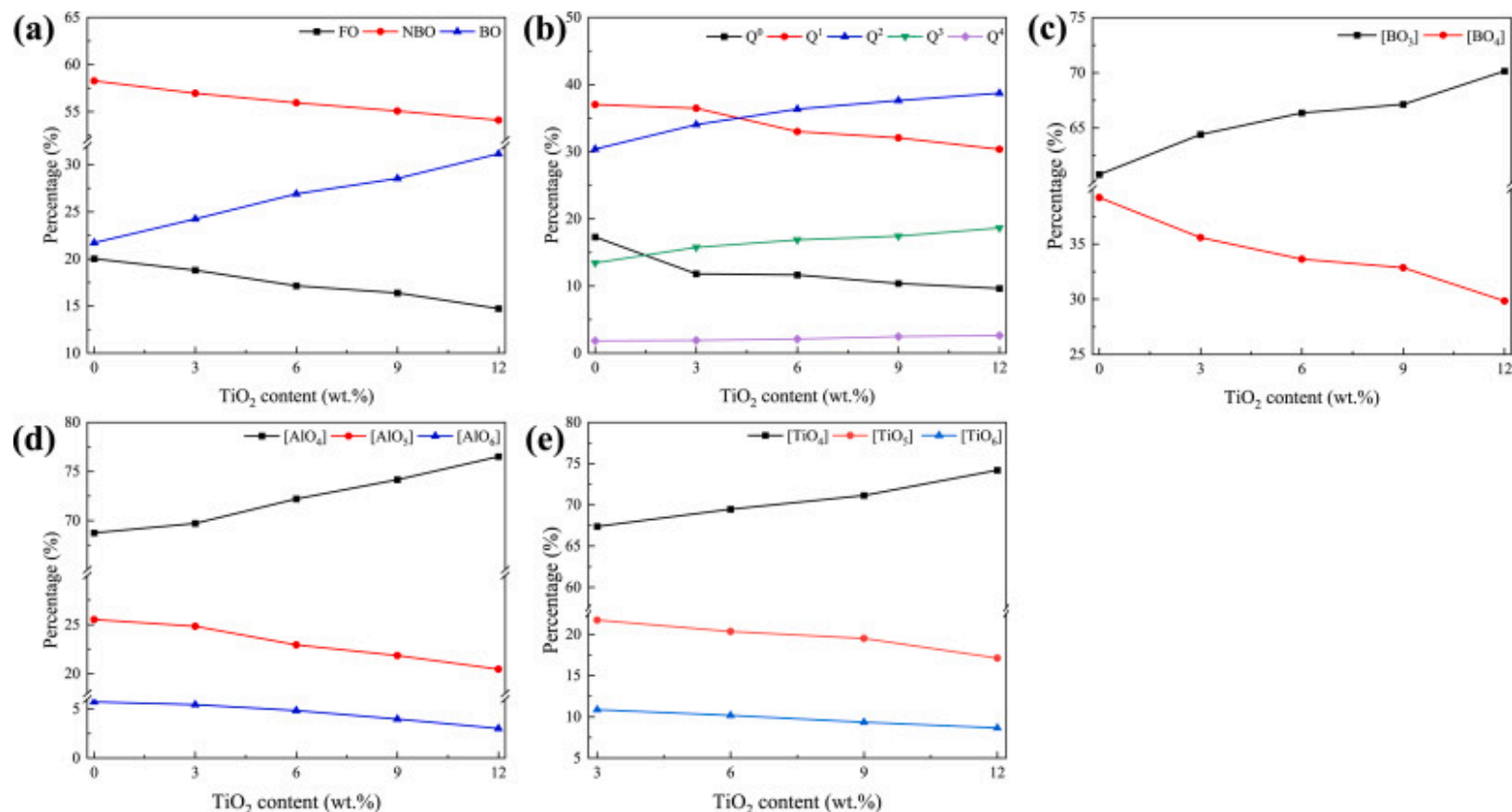
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Fig. 10. Results of XPS fitting for BSANBT glass at different TiO₂ contents: (a) Q⁰, Q¹, Q² and Q³; (b) [BO₃] and [BO₄]; (c) [AlO₄] and [AlO₆]; and (d) [TiO₄] and [TiO₆].

When TiO₂ content was increased from 0wt% to 12wt%, it can be seen that the percentages of Q³ and Q² were increased from 18.26% to 26.43% and 30.57%–38.89%, respectively, as shown in Fig. 10a. However, the percentages of Q¹ and Q⁰ were reduced from 33.60% to 24.24% and 17.57%–10.44%, respectively. Fig. 10b–d shows that the percentages of [BO₃], [AlO₄] and [TiO₄] were increased from 59.68% to 70.27%, 68.45%–76.64% and 66.80%–74.57%, respectively, with an increased TiO₂ content. This indicates that the evolution of BSANBT glass network structure gradually becomes complex with an increased TiO₂ content.

3.3.2. MD simulation analysis

Due to MD simulation can provide more information about the melt structure compared to spectroscopy techniques. The coordinate information for each particle at equilibrium, obtained from MATLAB ® software simulation, determines the amounts of bridging oxygen (BO, O⁰), non-bridging oxygen (NBO, O⁻) and free oxygen (FO, O²⁻) in the melt. The analysis also provides information on the amounts of B-, Si-, Al- and Ti-related units, as well as bond angles in the melt structure network, as shown in Fig. 11. It was found that BO amount was increased from 21.71% to 31.16% while NBO amount was decreased from 58.47% to 54.11% when TiO₂ content was increased from 0wt% to 12wt%, as shown in Fig. 11a. This indicates that BSANBT glass network DOP was gradually increased with an increased TiO₂ content. In addition, in order to describe complex BSANBT glass structures, (Q²+Q³+Q⁴)/(Q¹+Q⁰) amount ratios (R_{Si}), [BO₄]/[BO₃] amount ratios (R_B), [AlO₄]/([AlO₅]+[AlO₆]) amount ratios (R_{Al}) and [TiO₄]/([TiO₅]+[TiO₆]) amount ratios (R_{Ti}) are employed, as listed in Table 5.



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Fig. 11. Variation of structural unit percentage with TiO₂ content change: (a) FO, NBO and BO; (b) Q⁰, Q¹, Q², Q³ and Q⁴; (c) [BO₃] and [BO₄]; (d) [AlO₄], [AlO₅] and [AlO₆]; and (e) [TiO₄], [TiO₅] and [TiO₆].

Table 5. R_{Si} , R_B , R_{Al} and R_{Ti} of samples at different TiO₂ contents.

	0wt%	3wt%	6wt%	9wt%	12wt%
R_{Si}	0.84	1.07	1.24	1.36	1.5
R_B	0.65	0.55	0.51	0.49	0.43

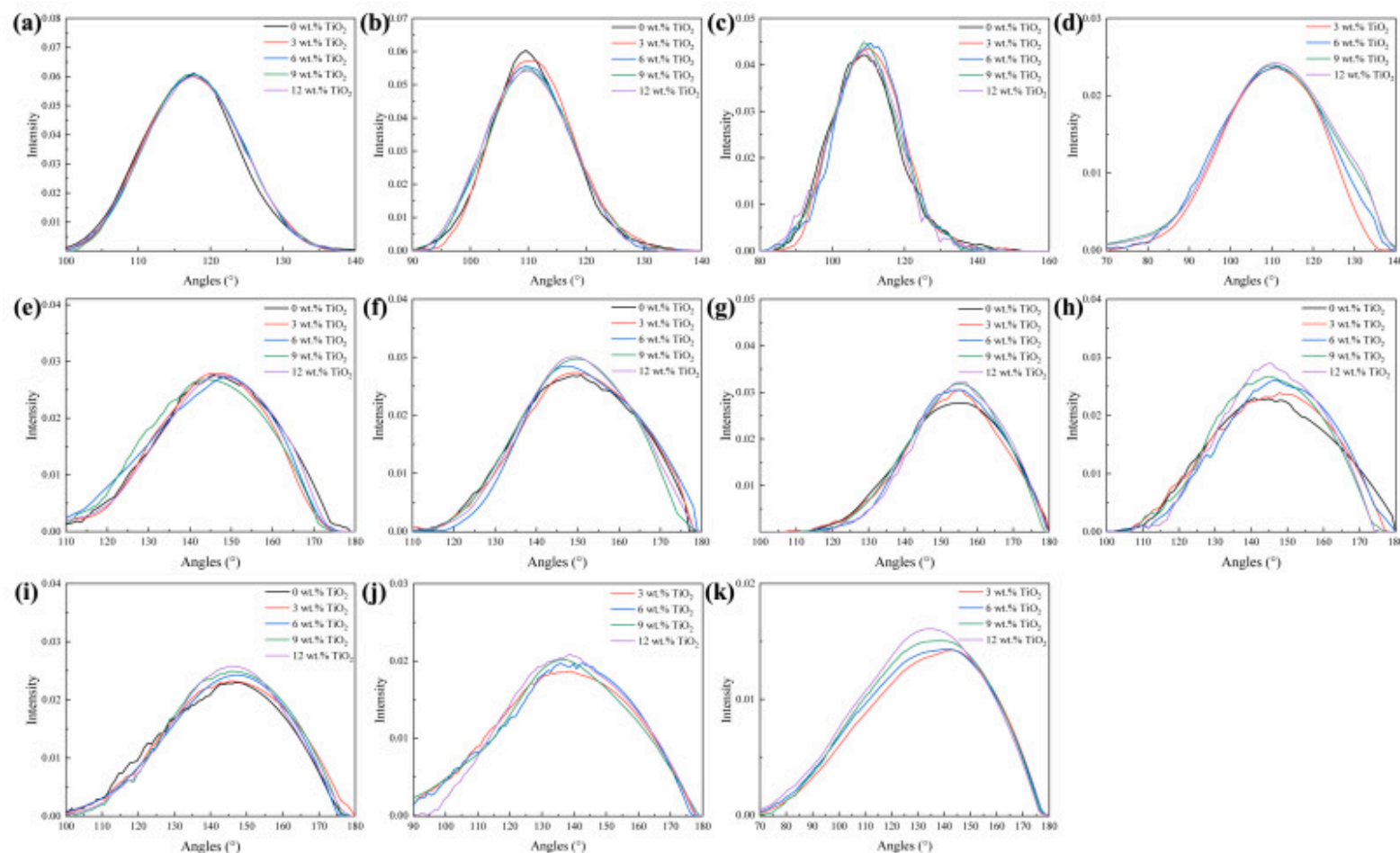
	0wt%	3wt%	6wt%	9wt%	12wt%
R_{Al}	2.20	2.30	2.60	2.87	3.26
R_{Ti}	–	2.07	2.28	2.47	2.88

Fig. 11b shows the variation of Si-related units with an increased TiO₂ content. It was found that the percentages of Q⁴, Q³ and Q² in the melt were increased, while the percentages of Q¹ and Q⁰ were reduced. These results are compared to Si 2p XPS spectral fitting results, which do not provide information on Q⁴ unit. This is due to the low intensity of the band energy of Q⁴ in XPS spectra of the mixed complex glass system [[56], [57], [58]]. Also, the percentage of Q⁴ was increased from 1.84% to 2.63% with an increased TiO₂ content. This indicates the Si-related unit information obtained from MD simulations is generally accurate. Furthermore, R_{Si} was increased from 0.84 to 1.5, when TiO₂ content was increased from 0wt% to 12wt%, which indicates an increased silicate network DOP.

The percentages of [BO₃] and [AlO₄] and the amount of [TiO₄] are increased with an increased TiO₂ content, while the percentages of [BO₄], [AlO₅], [AlO₆], [TiO₅] and [TiO₆] are decreased, as shown in Fig. 11c–e. The variations of [BO₃], [BO₄] and [AlO₄] percentages and [TiO₄] amount as a function of an increased TiO₂ content were found to be in good agreement with XPS spectral fitting results, while large differences exist in the percentages of [AlO₆] and [TiO₆]. This is due to the low intensity of the band energy of [AlO₅] and [TiO₅] in the XPS spectra of the mixed complex glass system [12,22,59,62,63]. This resulted in large errors in the percentages of [AlO₆] and [TiO₆] obtained from MD simulations and XPS spectral fittings, as shown in Fig. 11d and e. However, the percentages of [AlO₆] and [TiO₆] obtained from two analytical methods have the same trends with an increased TiO₂ content. In addition, R_B was reduced from 0.65 to 0.43 with an increased TiO₂ content from 0wt% to 12wt%, while R_{Al} was increased from 2.20 to 3.26, as shown in Table 5. R_{Ti} was also increased from 2.07 to 2.88 when TiO₂ content was increased from 3wt% to 12wt%. This indicates a gradually polymerized BSANBT glass network with an increased TiO₂ content, which is in the contrary to viscosity trend. In other words, the change of network DOP is not a major reason of a decreased viscosity. Furthermore, the mechanism of glass viscosity change due to an increased TiO₂ content is analyzed in the following sections.

Bond angle distributions (BAD) between cations in major glass network formers were calculated to understand the connections between cationic oxygen polyhedrons when TiO₂ content was increased. BAD of the BSANBT glass is shown in Fig. 12. The bond angles of O-B-O, O-Si-O, O-Al-O and O-Ti-O obtained from MD simulations were approximately 118.0°, 110.0°, 108.5° and 111.5°,

respectively, as shown in Fig. 12a–d. It was close to the previously reported BAD [40,64,65]. Because of the different coordination states of B and Si, the bond angles were approximately 147.5° for B-O-B and 150.5° for Si-O-Si, as shown in Fig. 12e and f, which is similar to the previous reports [[64], [65], [66]]. Additionally, Al-O-Al, Ti-O-Ti and Al-O-Ti were not observed in the melt network. It is due to that Al₂O₃ content was reduced from 4.28mol.% to 3.82mol.% when TiO₂ content was increased from 0wt% (0mol.%) to 12wt% (10.84mol.%), as shown in Table 1. This indicates that Al₂O₃ content is low in the BSANBT glass, which may results in Al-O-Al not being observed. Also, the atomic diffusion coefficients of Ti are smaller than those of Si, B and Al, as shown in Fig. 4b. This indicates that the Ti-O bond strength is weakest in the BSANBT melt and easily combines with B and Si atoms. It may result in Ti-O-Ti and Al-O-Ti not being identified in glass melts.



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Fig. 12. Bond angle distribution curves for each sample: (a) O-B-O; (b) O-Si-O; (c) O-Al-O; (d) O-Ti-O; (e) B-O-B; (f) Si-O-Si; (g) Si-O-B; (h) Si-O-Al; (i) B-O-Al; (j) Si-O-Ti and (k) B-O-Ti.

The BADs of Si-O-B, Si-O-Al and B-O-Al were distributed between 100° and 180°, as shown in Fig. 12g–i. The BADs of Si-O-Ti and B-O-Ti were divided between 90° and 180° and between 70° and 180°, respectively, as shown in Fig. 12j and k. Among them, the bond angles of Si-O-B, Si-O-Al and Si-O-Ti are about 151.5°, which are similar to those of Si-O-Si. This indicates that [SiO₄] dominates in the melt compared to B-, Al- and Si-related units, which agrees with previous simulations [66,67]. Furthermore, the bond angles of B-O-Al and B-O-Ti are approximately 148.0°, which are similar to those of B-O-B. It shows that B, Al and Ti-related structural connection processes are dominated by B-related units. Also, the bond angle strengths of B-O-Ti and Si-O-Ti are enhanced with an increased TiO₂ content. This indicates that the corresponding structure in the BSANBT glass increased as an increased TiO₂ content. However, the bond angle strengths of B-O-Ti and Si-O-Ti are lower than those of the other structures, as shown in Fig. 12, which shows fewer amounts of B-O-Ti and Si-O-Ti in the glass. Therefore, the above variations of BADs guide analyzing the evolution of the BSABNT melt structure due to TiO₂ content changes.

3.4. Mechanism of TiO₂ content addition on glass properties

The BSANBT glass lubricant should have suitable T_g and good wettability with the matrix. However, the viscosity was measured at a continuously decreased temperature, and T_g and the wettability between lubricant and matrix were increased during the analysis. Also, the melt structure information is obtained by holding the melt after cooling it to a specified temperature. Therefore, the correlation between melt structure and its viscosity is better. In this section, only the correlation between viscosity and structure was analyzed.

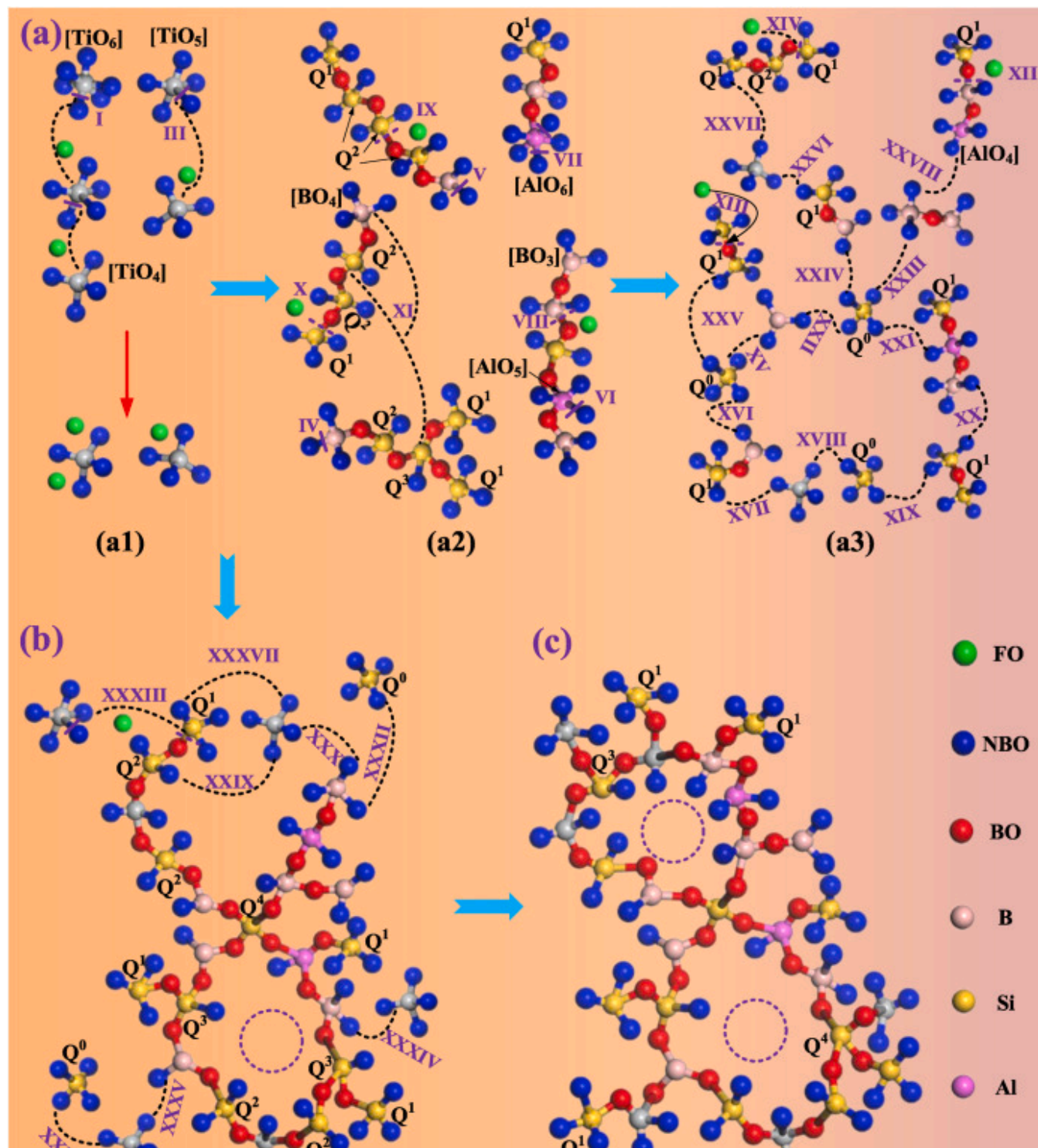
Na₂O and BaO act as network modifiers in BSANBT glasses to provide FO to the melt network [23,34]. (BaO+Na₂O) content was changed from 20.04mol.% to 17.87mol.% with a decrease by 10.83% when TiO₂ content was increased from 0wt% to 12wt%, leading to a reduced amount of FO. This resulted in a leftward equilibrium shift for Reaction (5), leading to an increased percentage of BO in the melt network, as shown in Fig. 11a. This in turn causes a further network polymerization. It also results in reduced percentages of Q⁰ and Q¹ in the melt network, while Q⁴ percentage is increased. This causes the equilibrium shift to the

rightward for the Reactions (6–7) [68], resulting in increased percentages of Q² and Q³ in the glass network, as shown in Fig. 11b.



A decreased amount of FO in the network when TiO₂ content was increased, resulted in a leftward shift of the equilibrium for the Reactions (8–9) [22,69]. This caused the transformation of [TiO₆] (as shown by I) and [TiO₅] (as shown by II and III) to [TiO₄], as shown in Fig. 13a1. The reduced amount of FO lead to an equilibrium shift to the left side in Reaction (10) [12] and Reactions (11–12) [22,69], promoting the transition of [BO₄] (as shown by IV and V) to [BO₃] and the transitions of [AlO₅] (as shown by VI) and [AlO₆] (as shown by VII) to [AlO₄], as shown in Fig. 13a2. At the same time, the combination of Q³, Q², and Q¹ with [BO₄] (as shown by XI) in the melt network also produces new [BO₃], as illustrated in Reaction (13) [70]. This also led to an increased percentage of [BO₃] in the melt network. Additionally, the generation of new FO during the above transformation of B-, Al- and Ti-related units disrupted the original Si-O-B (as shown by VIII and XII) and Si-O-Si (as shown by IX, X, XIII and XIV) structures and formed newly dissociated Si- and B-related units, which in turn are linked to [AlO₄] (as shown by XXI and XXVIII) and [TiO₄] (as shown by XVII, XVIII, XXVI and XXVII) to form complex glass structures, as shown in Fig. 13a3. Therefore, the above structure evolution indicates increased percentages of [BO₃] and [AlO₄] and increased amount of [TiO₄] in the network when TiO₂ content was increased.







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Fig. 13. Evolution of BSANBT glass melt structure with the change of TiO₂ content.

The network DOP was increased with an increased TiO₂ content, which may promote the formation of complex structures in the dissociated B-, Si-, Ti-, and Al-related units. This results in further increased amounts of BO, Q², Q³ and Q⁴, as shown in Fig. 12b. In addition, it is observed that a further increased TiO₂ content leads to more dissociation [TiO₄] and [TiO₅] in the melt network. Meanwhile, FO generated during the transformation of [TiO₅] to [TiO₄] destroys Si-O-Si (as shown by XXXIII) to form dissociated Q⁰ units. Also, Si-O bonding energy is stronger than B-O, while Ti-O has the lowest bonding energy in the melt. This facilitates the combination of dissociated Ti-related unit with B- and Si-related units to form new Si-O-B (as shown by XXII), Si-O-Ti (as shown by XXIX, XXXII, XXXVI and XXXVII) and B-O-Ti (as shown by XXX, XXXIV and XXXV). This results in decreased amounts of Q¹ and Q⁰ in the melt network, and the final formation of a complex glass network, as shown in Fig. 12c. Therefore, the symmetry and homogeneity of the glass network structure were weakened. In addition, the contents of SiO₂, B₂O₃ and Al₂O₃ were changed from 48.80mol.% to 43.51 mol.%, 26.88mol.% to 23.97mol.% and 4.28mol.% to 3.82mol.%, respectively, when TiO₂ content was increased. Also, the Ti-O bond strength was weaker than that of B-O and Si-O, as well as the contents of B₂O₃ and SiO₂ were decreased with an increased TiO₂ content. It indicated that [TiO₄] amount in the melt was increased, while the amounts of [SiO₄] and [BO₄] were decreased, which resulted in a weakened network strength. Meanwhile, R_B was decreased with an increased TiO₂ content, which indicates that the percentage of [BO₃] was increased, further weakening network strength. In a word, the network homogeneity and symmetry were weakened due to an increased TiO₂ content. This leads to a reduced viscosity in the temperature range of 950°C–1000°C.

Viscosity, T_s and wettability between lubricant and titanium matrix are important properties of glass lubricant. Previous studies [12,14] on the viscosity of glass lubricants in the hot working found that lubricants with a viscosity value of around 50Pas in the hot working temperature and a small change of viscosity with temperature were recommended. Therefore, lubricant T2 (25.50wt%B₂O₃-39.95wt% SiO₂-5.95wt%Al₂O₃-11.05wt%Na₂O-14.55wt% BaO-3 wt.%TiO₂) should be for the hot extrusion of TC4 alloys due to its small viscosity change in the temperature increase from 950°C (46.6Pas) to 1000°C (22.2Pas). Additionally, when T_s is low, the lubricant in contact with the titanium billet softens prematurely and is rapidly drained off under extrusion, resulted in a poor lubrication. For this reason, T_s should be generally 200°C–300°C lower than the hot extrusion working temperature [71]. This indicates that the BSANBT glass T_s was changed from 668.4°C to 700°C when TiO₂ content was increased from 0wt% to 12wt%, which meet the requirements of lubricants for TC4 alloy hot extrusion. Also, the minimum θ between lubricant T2 and TC4 alloy was 8.1°. This means that sample T2 has a good wettability on TC4 alloy. In summary, lubricant T2 was recommended for TC4 alloy hot extrusion in this study.

4. Conclusion

In this study, a novel glass lubricant of (26.29–0.263x)B₂O₃–(41.19–0.412x)SiO₂–(6.13–0.061x) Al₂O₃–(11.39–0.114x)Na₂O–(15–0.15x) BaO–xTiO₂ was proposed for the hot extrusion of TC4 alloys in the temperature range of 950°C–1100°C. Effect of TiO₂ content on viscosity, melting behavior, wettability and structure was investigated. The main findings are summarized as follows:

- (1) The glass viscosity shows a decreased trend with an increased TiO₂ content. The viscosity value at 950°C was decreased by 42.83% from 48.8Pas to 27.9Pas with TiO₂ content increased from 0wt% to 12wt%.
- (2) The percentages of Q⁴, Q³, Q², [AlO₄], [TiO₄] and [BO₃] in the melt were increased with an increased TiO₂ content, while the amounts of [SiO₄] and [BO₄] were reduced. This leads to a weakened network homogeneity and symmetry, leading to a reduced network strength, which determines viscosity changes.
- (3) The values of T_d , T_s , T_h and T_f of the lubricant were increased with an increased TiO₂ content. Furthermore, the lubricant has a good wettability with TC4 matrix at its T_{ht} , whereas lubricant T2 (25.50B₂O₃-39.95SiO₂-5.95Al₂O₃-11.05Na₂O-14.55BaO-3TiO₂) has the smallest wetting angle value (8.1°) on TC4 alloy matrix.
- (4) The lubricant T2 was recommended during the TC4 alloys hot extrusion for its T_s value of 759.1°C, viscosity value of 46.6Pas and wetting angle value of 8.1° at 950°C, as well as a small variation of the sample viscosity

in the temperature range of 950°C–1000°C.

CRediT authorship contribution statement

Lei Cui: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pei-yuan Ni:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Bing Song:** Writing – review & editing. **Wei Lv:** Writing – review & editing. **Ying Li:** Writing – review & editing.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this paper.

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